

Action-based Candidate Evidence (ACE)

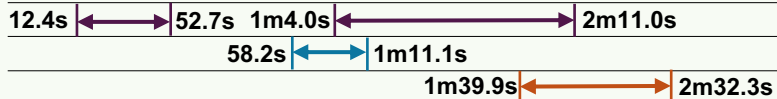
Temporal Verifier

Long
Input
Video



Action Detector

Read
Drink
Walk



LLM

Candidate 1: Walk
Candidate 2: Read
⋮

Evidence 1: 1m39.9s | 2m32.3s
Evidence 2: 58.2s | 2m11.0s
⋮

Query-Conditioned Proposal Generator

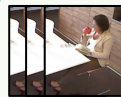
Query

“What action did the person perform while drinking something?”

Answer:
Read

Evidence Clip:

58.2s | 2m11.0s



Top
Candidate

Scoring &
Reranking

Vision Language
Model



Confident?

Answer: Read
Semantic Evidence: The scene shows a person reading a book while drinking from a red mug.

Visual Evidence:

